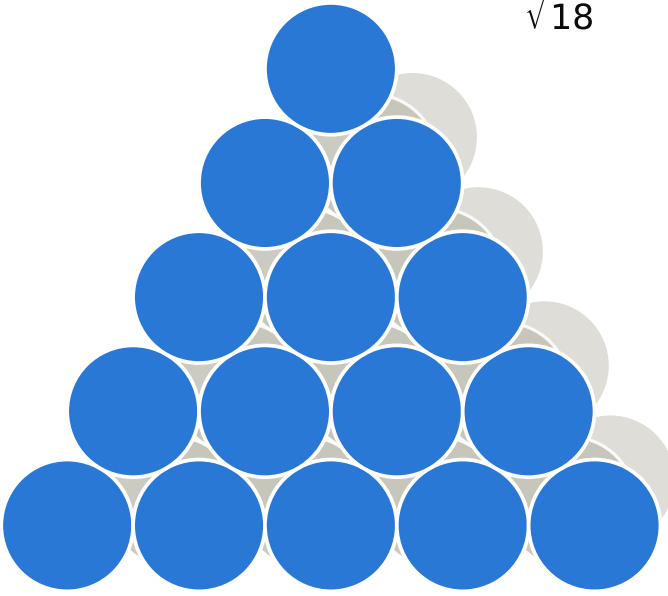
density $\frac{\pi}{\sqrt{18}} \approx 0.7405$



Kepler 1611: no packing of equal spheres is denser. Hales 1998, formally verified 2014 (Flyspeck).